



Time series data mining for railway wheel and track monitoring: a survey

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Abstract

The railway sector has witnessed a significant surge in condition-based maintenance, thanks to the proliferation of sensing technologies and data-driven methodologies, such as machine learning. However, despite the plethora of algorithms designed to detect and classify track irregularities and wheel out-of-roundness, they often fall short when put to the test in real-world scenarios. These shortcomings typically stem from their inability to meet all four critical requirements for constructing an effective maintenance plan: (R1) suitability of the condition-based maintenance strategy, (R2) availability of relevant data, (R3) proper problem formulation, and (R4) accurate evaluation of data mining methods. In response to the absence of a unified framework and standardized guidelines, this survey delves into the realm of time series sensor data and wheel-track interface components for railway structural health monitoring. This survey aims to bridge this gap by offering an extensive categorization, pinpointing existing challenges, and outlining potential directions for future research. Through these efforts, this survey provides a more thorough and targeted exploration of the subject matter, contributing to the advancement of this field.

Keywords Condition-based maintenance · Artificial intelligence · Railway infrastructure · Sensing technologies · Time series analysis

1 Introduction

The railway sector relies on reliability and maintainability to ensure the smooth and secure transportation of passengers and freight [1]. With trains designed to operate for decades and maintenance costs accounting for a significant portion of the total expenditure, there is an urgent need to keep safety and quality of service at an optimal level while

minimizing the operational costs. In fact, with high traffic levels, massive axle loads, and constantly changing conditions, even minor flaws in the railway track can be transformed into significant problems. In particular, the wheel-track interface has received a lot of attention in the literature, as it incurs most of the cost of maintenance for both railway vehicles and infrastructure [2]. Examples of such components include wheels, rails, sleepers, fastenings, and ballast, among others. These components are exposed to various environmental and operational conditions, which can induce corrosion, cracks, and other types of damage. In addition, surfaces of the wheel-track are subjected to high stick, sliding, and contact stresses in rolling contact, with the increasing hazard function being a common pattern observed in the failure rates of these mechanical components [3, 4]. To address this increasing probability of failure over time, it is imperative to perform maintenance actions proactively.

By implementing a preventive maintenance strategy, railway operators can reduce the likelihood of downtime, increase equipment lifespan, and avoid costly emergency

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repairs. The railway industry often uses a periodic replacement scheme based either on tonnage or kilometers traveled for vehicles and accumulated load or passing traffic for tracks [5]. Furthermore, these maintenance interventions are intuitively affected by expert knowledge on the effect of structural parameters on degradation rate. For example, change in rail profiles, closeness to switches and direct fastening systems increase the rate of degradation. In addition, the railway industry tends to have a comprehensive maintenance plan, by proactively lubricating, refurbishing, calibrating, tamping, and drive-by visual inspecting equipment on a regularly scheduled basis. However, these plans tend to be too conservative to compensate for the fact that they fail to tackle the specific root cause, which not only results in very high maintenance costs but can also reduce the component lifespan. For instance, it has been proven that despite tamping improves the condition of the track geometry, excessive ballast settlement can degrade the structural parameters of railway systems [6]. Ideally, the maintenance action should be performed just before the component failure. To tackle this challenge, the railway industry has witnessed the emergence of low-cost and easily maintainable condition monitoring systems, enabled by digital automation systems, Big Data, and Industrial Internet of Things (IoT) technologies. Both onboard and wayside condition monitoring are increasingly embraced with analysis tools to evaluate the railway load [7], as illustrated in Fig. 1.

The condition of a system is quantified by periodically or continuously collecting information from sensors mounted on its components, and actions are taken only when evidence of abnormal behavior is detected. This trend has given rise to an increasing use of artificial intelligence in condition-based maintenance, with numerous publications on this approach [8]. However, implementing data mining techniques for railway maintenance remains a challenge due to the absence of a common framework and standards [9]. For instance, many studies neglect to account

for the fact that deploying a greater number of sensors inevitably leads to increased costs. These costs arise from the need for additional equipment, maintenance, and data processing infrastructure. Therefore, a condition-based maintenance strategy is most advantageous for critical components whose failure does not occur instantaneously and can result in substantial function loss and safety risks. To address these issues, the survey investigates the challenges and opportunities of employing artificial intelligence for condition-based maintenance in railway systems, with a particular emphasis on the wheel-track interface and time series sensors. With this central research question in mind, the following specific questions (*SQs*) were posed:

(*SQ1*) What are the most common types of sensors used to collect time series data in railway systems, particularly for the wheel-track interface?

(*SQ2*) What are the most used machine learning algorithms for condition-based maintenance in railway systems, and how do they compare in terms of accuracy and computational efficiency?

(*SQ3*) What are the most effective preprocessing techniques for time series data in the context of railway maintenance, and how can they be optimized for different types of sensors and maintenance tasks?

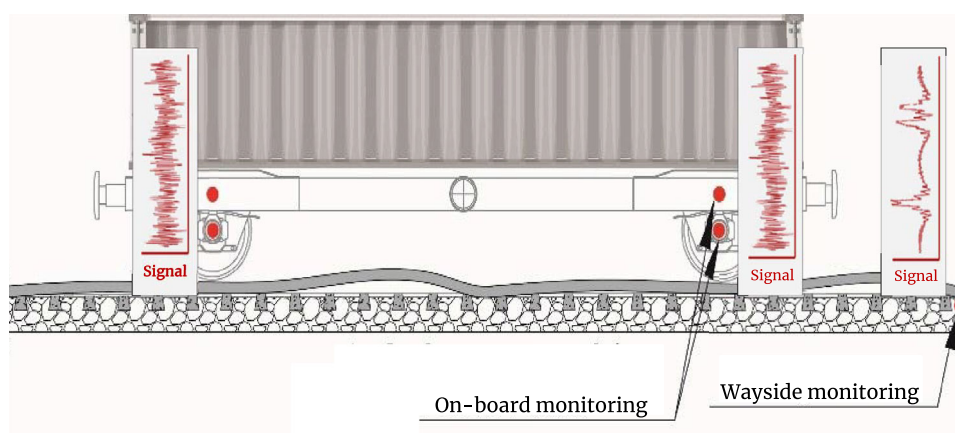
(*SQ4*) How are artificial intelligence-based maintenance models adapted to account for streaming data and automated maintenance decision-making?

(*SQ5*) How are computational expense, interpretability, and cost considerations incorporated into artificial intelligence-based maintenance models to improve trust and adoption by maintenance staff and decision-makers?

(*SQ6*) What are the key opportunities and challenges in scaling up the use of artificial intelligence for condition-based maintenance in railway systems, and what strategies can be employed to overcome these challenges?

These questions were addressed in the form of a comprehensive survey, which identified how the current

Fig. 1 Types of condition monitoring for proactive maintenance: onboard and wayside



literature complies with the key requirements (*R*) for creating an effective maintenance plan:

- (*R1*) Condition-based maintenance strategy is appropriate.
- (*R2*) Available data.
- (*R3*) Problem is appropriately formulated.
- (*R4*) Data mining methods are evaluated correctly.

As for *R1*, the mechanical components in the wheel-track interface present a gradual degradation process, e.g., wear and rolling contact fatigue, and a major threat to the operational safety of a railway system, thus condition-based maintenance is the right strategy [8]. Regarding *R2*, the dynamic context of the railway system is exceptionally challenging. Numerous measurement methods are used for bogies and tracks, including 3D-laser cameras, ultrasonic devices, eddy current, magnetic flux leakage, manual inspection, and others. As the data available impacts the selection and success of data mining methods, this survey focuses solely on time series, not only because it is the most common format of sensor data [7], but also due to the complexity of dealing with the sequential structure of data, with order and temporal dependencies. The most prominent problems arise from the high dimensionality and difficulty in devising similarity measures that consider temporal patterns, trends, periodicity, and other unique characteristics of time series data. Regarding *R3*, it is necessary to select effective preprocessing techniques to deal with the temporal dimension, such as for data representation, similarity measurement, and indexing. In addition, the modeling strategy can be posed as a fault diagnosis or prognosis task. Finally, for *R4* there are also significant gaps in evaluation standards, characterized by the absence of proper performance metrics, interpretability, cost

incorporation, and considerations for streaming environments [9, 10]. Figure 2 illustrates these requirements.

Within this context, this literature review poses the central question: What are the key challenges and opportunities in applying artificial intelligence to condition-based maintenance of railway systems, particularly in the context of the wheel-track interface, and how can they be addressed? To consider relevant papers, the choice of keywords for building the search strings was based on terms commonly found in the literature:

“PdM” OR “CBM” OR “diagnosis” OR “prognosis” OR “condition monitoring” OR “RUL” OR “PHM”) AND (“railway” OR “rail” OR “track” OR “wheel-track”) AND (“data-driven” OR “big data” OR “AI” OR “ML” OR “DL” OR “statistical”).

Note that this survey exclusively focused on primary research papers published in English between 2010 and 2023. Papers that didn’t use measurement systems related to time series or did not primarily focus on wheel and track condition monitoring were excluded from consideration. Following the stated search strategy, a total of 125 publications were thoroughly reviewed. Multiple databases were assessed during the research process, including ScienceDirect, SAGE, IEEE Xplore, Taylor & Francis, MDPI, and Springer. The survey is structured as illustrated in Fig. 3. Section 2 provides a comprehensive overview of related work, highlighting the main differences. Section 3 presents the parts and type of defects for which condition-based maintenance is an adequate strategy. Section 4 covers the kind of data being used, namely the measurement methods, typical characteristics of sensor data and support with physical-based models. Section 5 provides a thorough overview of the approaches developed to deal with time series sensor data, with different goals, learning tasks,

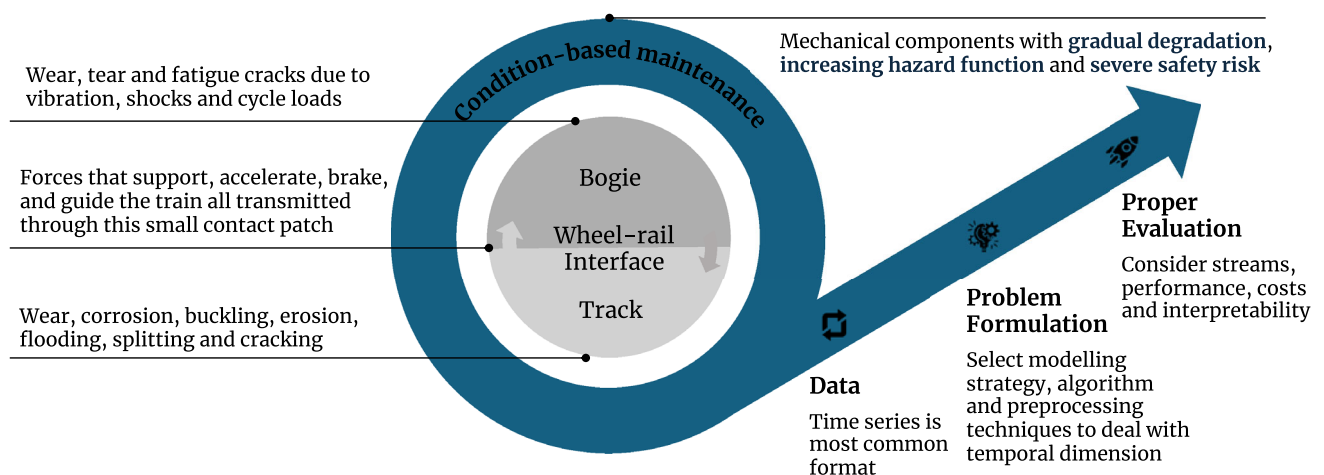


Fig. 2 An effective maintenance plan requires the maintenance strategy to be appropriate, available data, problem to be appropriately formulated, and data mining methods to be evaluated correctly

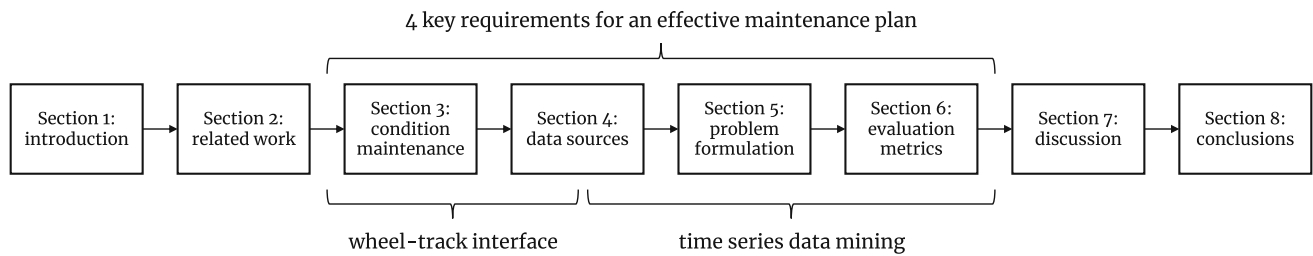


Fig. 3 Organization of the survey: eight sections

methods, and implementation components. Section 6 studies the most common evaluation procedures: performance, interpretability, cost, and computational expense. Finally, Sect. 7 links existing literature to potential research directions, and Sect. 8 concludes the survey.

2 Related work

In the context of prior research, early literature reviews focused on discussing different types of sensors, measurement objectives, and conditions. One notable contribution, in 2011, proposed a taxonomy that categorized monitoring systems into infrastructure-based infrastructure monitoring (I2I), infrastructure-based rolling stock monitoring (I2V), rolling stock-based infrastructure monitoring (V2I), and rolling stock-based rolling stock monitoring (V2V) [11]. Another comprehensive overview, in 2014, examined the use of wireless sensors for monitoring structures, vehicles, and machinery, shedding light on their practical applications [12]. However, the most widely adopted taxonomy formed the basis for a review conducted in 2017, which further classified monitoring systems into inspection and in-service categories, including onboard and wayside measurement [7]. In terms of wayside detection, monitoring systems were extensively reviewed, in 2005, focusing on vibration-based systems [6], and, in 2019, focusing on image processing for visual inspection [13]. In terms of in-service vehicles, a literature review, in 2015, addressed track geometry condition [14]. Furthermore, the literature has given considerable attention to wheel out-of-roundness. A comprehensive review, in 2000, provided a thorough analysis of the causes and modeling techniques of wheel out-of-roundness [15]. An extension to this review, in 2003, included causes of wheel corrugation and low-order polygonal wear, along with mitigation measures to tackle these challenges [16]. Additionally, in 2005, the effects and implications of wheel out-of-roundness on vehicle and track components were examined [17]. Since then, numerous studies have emerged over the past decade, continuously updating the state-of-the-art knowledge in this field [18]. Regarding methods for condition

monitoring, early surveys primarily focused on the application of physical-based degradation prediction models [19]. However, the advent of Big Data has led to a growing trend of using artificial intelligence for condition-based maintenance, with numerous publications exploring this approach. Despite these advancements, there remains a scarcity of surveys in railway maintenance that specifically focus on the application of artificial intelligence. Three notable surveys, in 2019, 2020, and 2021, addressed maintenance applications, measurement methods, data-driven models, and evaluation considerations, providing valuable insights into these areas [8–10]. However, these surveys covered a wide range of railway subdomains, sensor types, and failures, which somewhat limited their depth of analysis. In contrast, the survey presented in this paper aims to bridge the research gap by focusing on the formulation, development, and evaluation of data mining methods for condition-based maintenance, particularly in the context of time series sensor data and wheel-track interface components. By providing an extensive categorization, identifying open challenges, and outlining future research directions, this survey offers a more comprehensive and focused examination of the subject matter, contributing to the advancement of the field.

3 Condition-based maintenance

The first requirement for an effective maintenance plan driven by artificial intelligence is ensuring the adequacy of the condition-based maintenance strategy. This section specifically concentrates on the mechanical components in the wheel-track interface that present a gradual degradation process, e.g., wear and rolling contact fatigue, and a major threat to the operational safety of a railway system. Thus, ensuring condition-based maintenance is the right strategy [8].

In a railway system, the monitoring of the wheel-track interface involves multiple crucial components that require ongoing surveillance. Understanding the concept of failure and its associated failure modes is of utmost importance in this regard. Failure occurs when an item is no longer able

to perform its required function, resulting in a fault or state of inability, while each failure mode represents a distinct way the performance requirement fails. Given the complexity of the wheel-track interface and the diverse range of potential failure modes, this survey focuses on aggregating trends at the component level while highlighting the most common failure modes. These monitoring objects can be categorized into distinct areas of focus: wheel geometry, bogie structural components, track geometry, and track structural components.

Firstly, wheel geometry monitoring entails the assessment of essential wheel parameters such as diameter, flange height, and tread profile [4, 20–35]. By closely monitoring these geometric attributes, abnormalities can be promptly identified, enabling timely maintenance actions to be taken. One significant concern is the presence of out-of-round wheels. These include flats formed when a wheel set locks and skids along the rail, polygonization caused by repetitive loading and unloading cycles, and corrugated treads formed by excessive friction or contact with external objects. In addition to wheel geometry, monitoring the health and performance of other associated bogie components is vital. Among these components, the axle bearing serves a critical role by facilitating the transfer of loads and minimizing friction during the movement of the train [36–38]. Table 1 summarizes the reviewed works according to monitoring objects focusing on wheel geometry and bogie structural components.

Track geometry monitoring is a critical process that focuses on evaluating various irregularities that can have negative effects on train stability and passenger comfort. These irregularities include track misalignment, unevenness, and deviations from the desired rail geometric parameters [22, 39–67]. Key parameters of concern are longitudinal level for wear, cant for centrifugal force, twist for unevenness, alignment for lateral force, wide gauge, as well as other horizontal and vertical deformities. These failures typically initiate as small wear and cracks, but quickly propagate due to the substantial shear and normal stresses on the rail caused by rolling-sliding contact loading. As cracks progress, they can result in spalling, leading to material detachment from the rail surface.

Nonetheless, it should be noted that track geometry irregularities not only are responsible for train accidents,

but they can also lead to the birth of structural defects. These defects encompass several critical aspects, including the condition of joints, turnouts/switches, rolling element bearings, fasteners, sleepers, ballast, and subgrade. Joints are used to connect two rail sections together [68]. They provide flexibility to accommodate temperature changes and allow the rail to expand and contract without significant damage. Turnouts, also known as switches, enable trains to change tracks or move from one line to another [69, 70]. Rolling element bearings are used in various parts of the railway system to reduce friction and allow smooth rotation [37, 71, 72]. Fasteners, such as clips or bolts, are used to secure the rail to the sleepers that provide support and stability to the track [56, 73, 74]. The ballast, a layer of crushed stones or gravel, provides support, drainage, and stability to the track, while the subgrade refers to the underlying soil or formation on which the track is constructed [42, 75, 76]. The failure modes associated with these structural defects primarily involve wear, predominantly in curves, fatigue in the form of surface or subsurface cracks, and plastic flow resulting in rail corrugation. Table 2 summarizes the monitoring objects focusing on track geometry and track structural components.

The existing studies in this field have predominantly adopted a single-component perspective, which oversimplifies the modeling process by isolating interconnected system effects [77]. As a result, it often leads to inconsistent system-level conclusions. It is important to recognize that railway systems are complex and heterogeneous, composed of interdependent components such as rails, sleepers, joints and fasteners, ballast, and subgrade. Therefore, a more comprehensive approach is necessary, considering the multicomponent system dependence categorized into economic, stochastic, and structural dependence [78]. While a few studies have addressed the economic dependence aspect by exploring the cost benefits of simultaneous maintenance of multiple components [79–82], the literature remains largely insufficient in addressing both structural dependence, which involves maintenance connections between parts, and stochastic dependence, where the failure of one component affects the performance of other parts of the system. In fact, the prevailing multicomponent perspective found in the reviewed literature merely offers limited explanations of the dynamic

Table 1 Categorization of works based on monitoring objects: wheel geometry and axle bearing

Item	Failure modes	References
Wheel Geometry	Flats due to sliding and skidding, corrugated treads and polygonization with uneven wear, material fallout, such as shelling or spalling	[4, 20–35]
Axle bearing	Material loss due to wear, inadequate lubrication, and fatigue	[36–38]

Table 2 Categorization of works based on monitoring objects: track geometry, joint, turnout, rolling bearing, ballast, and fastener

Item	Failure modes	References
Track Geometry	Rail head, breaks, longitudinal level, twist, and cant	[22, 39–67]
Joint	Misalignment and wear	[68]
Turnout	Misalignment	[69, 70]
Rolling bearing	Wear and overheating	[37, 71, 72]
Fastener and Sleeper	Corrosion and exposure to moisture	[56, 73, 74]
Ballast and Subgrade	Settlement and deformation	[42, 75, 76]

interaction between rolling stock and track components. Some studies focus on examining the interaction between friction on sharp curves, gauge corners, and track conditions with defects [83–86]. Others investigate the impact of overloads, weight-in-motion, and unbalanced loads on the rail [87–90]. Furthermore, recent works have utilized a simplified track quality index that integrates various factors, including dynamic effects, speeds, loads, and rail geometry parameters [39, 58, 91, 92], and key features of rail geometry parameters, such as gauge and twist, represented with standard deviations [57, 58, 93].

4 Data sources

The second requirement for an effective maintenance plan driven by artificial intelligence is addressing the characteristics of the available data in the challenging dynamic context of the railway system. To make optimal maintenance decisions, a vast amount of dynamic data from different sources is required, including service failure data, signal data, ballast history, grinding history, remedial action history, traffic data, and inspection data. As the data available impacts the selection and success of data mining methods, this survey focuses solely on time series, not only because it is the most common format of sensor data [7], but also due to the complexity of dealing with the sequential structure of data, with order and temporal dependencies. There are four main measurement methodologies: manual inspection, dedicated track inspection vehicles, onboard systems, and wayside systems. Different sensors are used depending on the methodology. This survey specifically concentrates on sensors that capture time series data, such as eddy current, vibration, displacement, impact load detectors, optical fiber, and ultrasonic sensors.

4.1 Measurement techniques

Manual inspection involves a walking patrol looking for signs of track defects. Onboard techniques involve installing sensors on the vehicle to monitor the condition of the track and the wheels themselves [57, 94, 95]. These can be

mounted on in-service vehicles that provide real-time data for long lengths of track [36, 51, 52, 62, 84–86, 96–102]. Alternatively, track inspection vehicles are normally carried out periodically and rarely on busy routes [21, 22, 48, 49, 53, 54, 103, 104]. In addition, wayside measurement systems allow the extraction of significant amounts of data referring to all the operating vehicles with a reduced number of sensors [23, 24, 83, 87, 91, 92, 105–109]. Table 3 highlights the four measurement techniques.

Within these measurement techniques, various sensors are available to efficiently monitor and control different process parameters, including force, machine vision, optical geometry, acoustic, fiber Bragg grating, pressure transmitter, eddy current signal, ultrasonic, magnetic, and thermal. Force measurement detectors are widely used to monitor the impact between wheels and the track structure. These detectors employ vibration-based systems to accurately capture the forces involved [36, 37, 51, 83, 93, 102, 110–112]. Vertical loads are typically measured in zones where the train is not accelerating or braking, while lateral loads are best measured in narrow curves. A commonly used sensor is the wheel impact load detector, which weighs each wheel multiple times and provides dynamic impact load information at the wheel level [36, 37, 51, 83, 93, 102, 110–112]. In addition, many systems deal directly with acceleration data [43, 56, 62, 68, 113, 114]. Machine vision technology utilizes advanced computer algorithms to process digital image data obtained from vehicle's underframes and side frames [13, 91]. By analyzing this information, machine vision systems can capture various wheel features, e.g., flange height, flange thickness, rim thickness, and diameter, as well as the condition of vehicle components. Laser-based systems also play a significant role in track and wheel monitoring [24, 58, 60, 91]. For example, optical geometry detectors employ cameras mounted on tangent tracks to calculate various parameters for each set of axles based on the angle of attack and tracking position. These systems enable precise measurements of the track's elevation, curvature, and alignment, which are crucial for ensuring safe and efficient railway operations. Longitudinal level, a commonly measured parameter, helps maintain track stability and smoothness [55, 75, 104, 115]. Acoustic

Table 3 Categorization of works based on measurement: manual inspection, inspection vehicle, in-service vehicles, and wayside sensors

Method	Description	References
Manual inspection	Walking patrol looking for signs of potential failure	[57, 94, 95]
Inspection vehicles	Mechanized track patrol carried out with sparse frequency	[21, 22, 48, 49, 53, 54, 103, 104]
In-service vehicles	Onboard sensors providing real-time data	[36, 51, 52, 62, 84–86, 96–102]
Wayside sensors	Fixed sensors providing real-time data	[23, 24, 83, 87, 91, 92, 105–109]

emission monitoring involves processing the unique noise signatures emitted [23, 116, 117]. By analyzing these acoustic signals, anomalous patterns can be detected. In addition, ultrasonic sensors are extensively employed in track and wheel monitoring applications [21, 44, 59, 61, 85, 103, 108, 109]. They use ultrasonic waves to measure parameters such as distance, thickness, and defects in materials. Fiber Bragg grating sensors are widely utilized in track and wheel monitoring due to their immunity to electromagnetic interference, multiplexing capabilities, long reach, lightweight nature, and high signal fidelity [20, 87, 92, 106, 107, 118]. These measure parameters such as strain [43, 53, 54, 57, 83, 84], temperature [36, 96], and pressure [91]. Eddy current testing involves inducing electrical currents in conductive materials to evaluate their integrity and detect flaws or discontinuities [44, 59, 63, 69]. Magnetic are also used to detect the changes in the signature of metallic objects [22, 86].

4.2 Type of data

Furthermore, combining contextual data, e.g., maintenance inspection history and traffic volume, with sensors is crucial for more accurate analysis, trend identification, and decision-making [39, 59, 63, 91, 104, 119, 120]. The use of both real and simulated data is prevalent in railway condition monitoring. Simulated data provides researchers with several benefits [4, 26, 29–32, 86, 90, 110, 120, 121]. One notable advantage lies in the ability to conduct controlled and repeatable experiments, which empower researchers to manipulate variables and observe their impacts without the constraints of real-world data collection. Nevertheless, it is crucial to acknowledge the limitations and the necessity for validation through field trials using authentic data. Field trials offer invaluable insights into the performance, reliability, and generalizability of models, algorithms, and interventions [36, 39, 40, 43, 53, 54, 58, 66, 91, 96, 97, 105, 106, 114, 115, 119, 122]. They enable researchers to identify potential issues, assess the influence of confounding factors, and uncover unexpected outcomes. In fact, the current state of research still demonstrates an absence of algorithms that have been

validated using real streaming data. Some efforts have been made to address this challenge by developing algorithms that perform computationally intensive training offline, resulting in models ready for rapid real-time predictions [20, 36, 49, 56, 91, 92, 96, 100, 101, 123, 124]. Table 4 presents an overview of these data types, highlighting the absence of practical implementations of complete online algorithms.

5 Problem formulation

The third requirement for an effective maintenance plan driven by artificial intelligence is to select effective pre-processing techniques to deal with the high dimensionality and temporal dimension, such as for data representation, similarity measurement, and indexing. In addition, it is crucial to consider not only the available data but also the desired output that the model should provide, and time-frame involved. To illustrate this process, let's consider the example of rail wear caused by the friction between train wheels and the tracks. It is important to formulate specific goals based on the business requirements or the maintenance staff needs. For instance, are there requirements to detect rail sections with an uneven longitudinal level in real-time? Or perhaps the goal is to predict sections nearing the end of their useful lifespan and schedule replacements for the next year, treating it as a classification problem? Another approach could involve survival analysis to predict the probability of the contact rail having a level below a specific threshold within the next three years. Alternatively, the goal might be to forecast the longitudinal level in one year's time. Once the problem formulation aligns with the available data, it becomes crucial to select appropriate data mining methods as well as key implementation components representative of time series data. For example, if the focus is primarily on condition monitoring, a state space model might be adequate for capturing the dynamics of the system. However, if there is a lot of contextual information available, other machine learning models such as neural networks, decision trees, or support

Table 4 Categorization of works based on characteristics of data: simulated, real, contextual, and stream

Type	Description	References
Simulated data	Controlled and repeatable experiments	[4, 26, 29–32, 86, 90, 110, 120, 121]
Real data	Acknowledge all uncontrollable interferences	[36, 39, 40, 43, 53, 54, 58, 66, 91, 96, 97, 105, 106, 114, 115, 119, 122]
Contextual data	Failures, ballast history, traffic, inspections, and interventions	[39, 59, 63, 91, 104, 119, 120]
Data stream	Recognize high-rate frequency of continuous flow of data	[20, 36, 49, 56, 91, 92, 96, 100, 101, 123, 124]

vector machines might be better suited to handle the multidimensionality.

5.1 Modeling strategy

Predictive maintenance involves six modeling strategies, serving distinct purposes and relying on specific types of data. These strategies can be grouped into fault diagnosis and fault prognosis. When a machine experiences a breakdown, a substantial amount of time is typically spent on pinpointing the causes of the failure, while only a minor portion is allocated to the actual repair.

Fault diagnosis involves identifying the factors responsible for the deterioration observed in a component with three stages involved: detection, isolation, and identification [4, 20, 21, 25, 26, 29–32, 37, 42, 50, 57, 68–70, 73, 74, 94–96, 100, 107, 109, 111, 112, 116, 121, 123–132]. Detection involves anomaly detection and benchmarking machine behavior to identify deviations. Isolation determines the specific kind, location, and time of fault occurrence, such as identifying the exact component of the track, bogie, or defective wheel. Identification focuses on quantifying the size and behavior of the fault. Fault prognosis aims to answer questions related to the timing and extent of failure or degradation in the observed equipment, in three different ways: failure prediction, survival analysis, and remaining useful life (RUL) estimation. Failure prediction involves classifying failures within a specified time window, which helps optimize maintenance strategies [34, 39, 41, 47–49, 60, 91, 98, 101, 104, 105, 110, 114, 119]. Survival analysis is used to assess the probability of failure over time, providing insights into component reliability and longevity [23, 24, 83, 87, 91, 92, 105–109]. Finally, it is possible to estimate the remaining useful life of the components [36, 40, 41, 43, 45, 49, 51, 52, 54, 62, 64, 72, 84–86, 96–102, 118, 120, 133]. Table 5 highlights several works with fault diagnosis and prognosis modeling strategies.

5.2 Learning method

The modeling strategy highly influences the learning method of choice. For failure diagnosis, the most common are query by content which involves measuring the similarity of individual instances against a normal baseline [132], anomaly detection which involves ranking a set of instances, from the most anomalous to the most normal [96, 121, 126] and clustering by finding intrinsic groups of types or severities of damage [44, 128, 129]. While these techniques provide essential avenues for failure diagnosis without the need for labeled data, their unsupervised nature poses inherent limitations. Without access to ground truth labels, these techniques struggle to distinguish outliers or anomalies accurately, leading to false positives and false negatives. Moreover, their reliance solely on intrinsic dataset properties makes them sensitive to changes in data distribution over time, hindering their ability to adapt and accurately identify deviations from normal behavior. Additionally, clustering algorithms may encounter challenges when dealing with high-dimensional or nonlinear data. In this regard, classification models have indeed demonstrated superior performance [45, 119]. Furthermore, their use enables the extension to the more enriching modeling strategies for failure prediction and RUL estimation. By leveraging labeled data, these techniques can not only accurately identify failure instances but also provide valuable insights into the progression of failures over time. This allows for the development of more sophisticated predictive models that consider various factors influencing failures, such as temporal patterns, environmental conditions, and operating parameters. However, despite this potential, traditional algorithms continue to dominate the literature and often lag the state of the art, indicating that the full realization of these advantages has not yet been achieved. For example, kNN struggles with computational complexity and sensitivity to irrelevant features [97, 101, 134], while decision trees are prone to overfitting and instability [91, 97, 134]. Ensemble methods like random forests [43, 61, 97, 119] and AdaBoost [73, 74]

Table 5 Categorization of works based on modeling strategies: fault diagnosis, failure prediction, RUL estimation, survival analysis

Measurement	Problem framing	References
Fault Diagnosis	Is there a potential failure? Where? What? How severe?	[4, 20, 21, 25, 26, 29–32, 37, 42, 50, 57, 68–70, 73, 74, 94–96, 100, 107, 109, 111, 112, 116, 121, 123–132]
Failure prediction	Will a failure happen in the next year?	[34, 39, 41, 47–49, 60, 91, 98, 101, 104, 105, 110, 114, 119]
RUL Estimation	When will the next failure happen?	[36, 40, 41, 43, 45, 49, 51, 52, 54, 62, 64, 72, 84–86, 96–102, 118, 120, 133]
Survival analysis	What is the probability of failure over time?	[23, 24, 83, 87, 91, 92, 105–109]

enhance classification accuracy but are sensitive to noisy data and outliers. Logistic regression assumes linear relationships and may underperform with nonlinear data [39, 46, 121]. Support vector machines' performance depends on kernel and parameter selection, making them less suitable for large datasets and noisy environments [37, 43, 51, 58–60, 69, 91, 94, 101, 120, 134]. To overcome these limitations and push the boundaries of failure diagnosis and prognosis, recent efforts have explored novel benchmark models. For instance, deep learning architectures like convolutional neural networks (CNNs) and recurrent neural networks (RNNs) offer the capability to handle complex, high-dimensional data, and capture temporal dependencies effectively [57, 100, 101, 110, 111, 120]. Moreover, Bayesian methods present a promising avenue for enhancing failure diagnosis tasks [42, 63, 101, 104, 121]. Unlike traditional methods that often struggle with uncertainty estimation, Bayesian approaches offer a principled framework for probabilistic inference, allowing for robust uncertainty quantification in predictions. By explicitly modeling uncertainty, Bayesian methods can provide more reliable assessments of failure likelihoods, particularly in the railway setting with limited or noisy data. Additionally, Bayesian techniques facilitate the incorporation of prior knowledge or domain expertise into the modeling process, enabling more informed decision-making and improved generalization performance. Lastly, for survival analysis both probability distribution models [63, 75] and stochastic processes [72, 76, 115] are used in incorporating time-related variables and

considering the dynamic nature of failures. Table 6 encapsulates the link between common learning methods and tasks.

5.3 Implementation components

When analyzing time series sensor data, it is also important to consider several factors that can impact the success of data mining methods. Noise, inconsistencies, missing, or duplicated data are common challenges in railway condition monitoring with low-cost sensors. Fluctuations in reported values may result from minor variations in sensor sensitivity, unrelated events near the sensor, or transmission errors. To circumvent these, the fundamental shape characteristics of the time series signals are typically extracted through three groups of techniques: shape-based, feature-based, and model-based [135]. For short time series, shape-based representation is the most used, while for long time series both feature-based and model-based representations are derived and compared. Furthermore, these techniques are often accompanied by various signal pre-processing stages. The most common are noise filtering, scaling differences, and contextualization [10]. Noise filtering is handled with traditional signal processing techniques like digital filters or wavelet thresholding. Resampling is also typically used to smooth the noise in high-frequency signals. Scaling differences between time series are handled with normalization and transformation invariances. Contextualization is handled with data fusion techniques. This need for contextualization comes from the fact that sensors are distributed in the railway environment.

Table 6 Categorization of works based on learning methods: query by content, anomaly detection, clustering, classification, forecasting, regression, probability distribution model, and stochastic process

Learning task	Common learning methods	References
Fault diagnosis	Query by Content, anomaly detection, clustering	[44, 96, 121, 124, 126, 128, 129, 132]
Failure prediction	Classification	[39, 41, 60, 91, 98, 101, 105, 110, 114, 119]
RUL Estimation	Forecasting, regression	[36, 40, 41, 43, 45, 49, 54, 64, 72, 97, 118, 120, 133]
Survival analysis	Probability distribution model, stochastic process	[63, 72, 75, 76, 115]

On the one hand, wayside sensors present spatial relationships which indirectly determine a temporal correlation between observations. On the other hand, onboard sensors are consistently facing dynamic contexts of weather, soil type, and others. To circumvent this problem, the most used technique is principal component analysis [4, 26, 45, 58, 60, 95]. Also, a recent work proposed deep autoencoder to capture and compress the salient features of the input data [45].

Feature-based is the most used approach to represent the fundamental shape characteristics of a time series in wheel-track condition monitoring. Various advanced signal processing methods have been proposed that extract condensed damage-sensitive knowledge, with both statistical and frequency-domain features, e.g., the Fourier transform [52, 85, 113]. However, time–frequency techniques have been the most widely used, due to their ability to identify content of a signal in the frequency-domain without losing information about its time domain characteristics [84, 102, 105, 113]. Three commonly used techniques are: the Wavelet Transform [37, 50, 56, 58, 62, 84, 99, 102, 111, 130], Wigner-Ville Transform [84, 102], and Short-time Fourier Transform [4, 84, 102]. Furthermore, other approaches were developed to help dealing with the nonlinear effects on the suggested time-dependent characteristics. For instance, changes in train speed tend to affect the impact load, but this can vary based on the specific length of wheel flats, not following the pattern of increased impact load with higher train speeds [136]. Some alternatives were then suggested to extract the features, such as a time–frequency kurtosis to reduce surrounding noise and highlight faulty signal patterns of wheel flats [116], spectral kurtosis [31], symbolic data [131], and empirical mode decomposition to transform the signal into several intrinsic mode functions that isolate the failure signal mode from interferences and a criterion based on signal lag and envelope spectrum [29, 30].

Model-based representation consists in directly describing the dynamic behavior of the signal, such as autoregressive models [33, 64, 115], state space models, e.g., the Hidden Markov Model [34, 35, 38, 41, 42, 48, 55, 133] and regression [43, 134]. In

fact, model-based representations are also frequently used to extract coefficients which are then used in a feature-based approach [75, 91, 97, 110]. Recently, there has also been an increased use of deep learning models. Recurrent neural networks (RNNs) have become popular architectures used for railway condition monitoring, including long short-term memory (LSTM) for capturing long-term dependencies in sequential data [40, 43, 53, 119]. Alternatively, convolutional neural networks (CNNs) despite commonly associated with image recognition tasks, also have been applied to time series data with some modifications [68, 112].

Shape-based representation directly focuses on distinguishing or matching any pair of time series with an intuitive distance. To formalize this measure, it is necessary to establish a notion of similarity based on perceptual criteria rather than strict mathematical identity. In railway condition monitoring, dynamic time warping (DTW) [70, 127, 129, 132] has been preferred over the Euclidean distance [125], as it allows for the recognition of perceptually similar objects, even if they are not identical in mathematical terms. Table 7 categorizes these algorithms into the three described groups.

Finally, the stationarity of time series data must be considered. A stationary time series has consistent mean, variance, and autocorrelation over time. However, IoT data often exhibit non-stationarity in the form of concept drifts, seasonality and change points. Concept drift refers to changes in the statistical distribution of data over time, while seasonality involves cyclical changes occurring over longer time scales. Change points represent abrupt and permanent shifts in the normal state of a monitored system. The standard solution involves the identification of change points, then determining functions for curve fitting the intervals between them [27, 36, 137].

6 Evaluation metrics

The final requirement for an effective maintenance plan driven by artificial intelligence is to properly evaluate the devised models for performance, interpretability, cost

Table 7 Categorization of works based on implementation components for time series: feature-based, model-based, and shape-based

Measurement	Problem framing	Works
Feature-based	Statistical, time–frequency, spectral, model coefficients	[4, 29–31, 50, 52, 56, 58, 62, 84, 84, 84, 85, 99, 102, 102, 102, 111, 113]
Model-based	Autoregressive, HMM, Regression, RNN, LSTM	[33–35, 38, 40–43, 48, 52, 55, 64, 97, 115, 133]
Shape-based	Euclidean, DTW	[70, 125, 127, 129, 132]

incorporation, and considerations of streaming environments. Assessing the predictive capabilities of an intelligent system for determining when train maintenance is needed extends beyond mere accuracy evaluations. It entails a comprehensive evaluation encompassing various facets. These include performance assessment, consideration of cost implications associated with false positives and false negatives, the interpretability of the model, and continuous monitoring of the system's performance through real-time data analysis. Moreover, datasets related to rail defects often exhibit a highly imbalanced distribution, with a substantial majority of observations falling within the non-defective category. This skewing significantly challenges the reliability of standard evaluation metrics. Therefore, the selection of an appropriate evaluation metric becomes pivotal. In this regard, three primary categories of metrics should be considered: threshold metrics, ranking metrics, and probability metrics [48, 91, 96, 97, 110, 121, 132, 138]. Threshold metrics, such as accuracy, sensitivity–specificity, and precision–recall, quantify classification prediction errors. Ranking metrics evaluate classifiers based on their ability to separate classes and include the ROC curve and precision–recall curve. Probabilistic metrics quantify uncertainty in a classifier's predictions, which is useful for penalizing wrong but highly confident predictions. Intrinsically related to this imbalanced distribution is the notion of a false alarm or false safe prediction. From the engineer's perspective, high false alarm prediction usually leads to ineffective and unnecessary decision-making, while false safe prediction can cause huge loss for the railway service suspensions. While it is crucial to customize these definitions to suit the specific needs and data characteristics of a business, there has been a limited exploration of the economic implications stemming from the performance of these proposed methodologies [65–67, 122]. To compute these costs accurately, it is imperative to consider mandatory regulations, operational imperatives related to equipment uptime and ensuring the safety of personnel, as well as the constraints associated with the availability of maintenance staff and track infrastructure.

Additionally, in most studies, a significant challenge emerges in bridging the gap between evaluating sensor

types, their placement, and repair strategies, all while accommodating the need for human operators to receive assistance from AI in managing multi-objective trade-offs and addressing diverse priorities [139]. This inherent gap in the problem formulation is unavoidable, stemming from the nuanced dynamics of these trade-offs, necessitating case-specific decision-making. To address this inherent incompleteness, the incorporation of explanations becomes pivotal in ensuring the trustworthiness of the decision-making process [48, 55, 75, 83, 91, 95, 96, 119]. This includes developing visual analytics techniques, prototypical examples, and deductive argumentative systems for explanations. Finally, the large volumes of sensor data impose on algorithms memory and time constraints. Thus, algorithms must be evaluated on scalability and accuracy over time in data stream settings [91, 132, 140]. Table 8 categorizes these metrics into the four described groups.

7 Discussion

Despite the substantial body of work dedicated to this field in recent years, the research seems to have been predominantly motivated by the exploration of novel approaches, rather than a direct response to practical issues. In fact, there is a lack of robust business-oriented justifications for problem formulation, temporal dimension detailing, and technique selection within current practices. To address this gap, forthcoming research should delve into methodologies explicitly integrating business objectives into the analytical pipeline, in light of the four discussed requirements for creating an effective maintenance plan: condition-based maintenance strategy appropriateness, data source, problem formulation, and complete evaluation.

This detachment from practicality in the literature is exemplified by the neglect of more complex fault diagnosis tasks, such as isolating specific types, locating occurrences, and accurately quantifying fault behavior. Additionally, fault prognosis has tended to prioritize timing over assessing the full extent of failure or degradation. There is a pressing need for literature to move beyond mere anomaly detection and single-component behavior benchmarking to identify peak deviations. To tackle these

Table 8 Categorization of works based on evaluation metrics: performance, interpretability, costs, and computational expense

Evaluation	Metric	References
Performance	Threshold, ranking, probabilistic	[48, 91, 96, 97, 110, 121, 132, 138]
Interpretability	Visual analytics, prototypical examples, deductive argumentative systems	[48, 55, 75, 83, 91, 95, 96, 119]
Cost	Mandatory regulations, operational imperatives	[65–67, 122]
Computational expense	Memory requirements, computational time, stream	[91, 132, 140]

challenges, three distinct research directions emerge from the surveyed papers: uncertainty quantification, explanation techniques, and multicomponent modeling.

Quantifying model uncertainty is key in railway environments where the cost of a mistake can be huge, allowing for safer actions and human interventions. In this regard, leveraging the surveyed techniques on probabilistic modeling and Bayesian methods already holds promise for quantifying uncertainty. However, these measures were estimated on the whole testing data, which may introduce biases, akin to the challenges encountered by dealing with imbalanced classes. The uncertainty associated with the typically small groups of samples associated with failure could potentially become biased toward the performance on the rest of the data. Thus, future research lies in developing pointwise evaluation measures to assess uncertainty accurately. Indeed, these estimates would not only serve as crucial indicators but also provide explanations for the model's predictions. This is pivotal in ensuring trustworthiness and facing the inherent formulation incompleteness of a condition-based maintenance problem.

Alternatively to uncertainty estimates, relevant and nonredundant feature subsets can be detected through interpretable models or post hoc explanation techniques. Interpretable models, such as sparse neural networks, induced by structured weight pruning and group sparsity regularization, offer transparency by directly showing how each feature contributes to the prediction outcome. Post hoc explanation techniques, such as layer-wise relevance propagation and feature visualization methods, by analyzing the patterns that maximally activate individual neurons or layers, provide insights into which parts of the input contribute most to the model's decision. Furthermore, ensuring the stability of these explanations over time is crucial for building trust. Thus, an open challenge in railway maintenance still lies in computing multiple model explanations over time, tracking attribute-value contributions for prediction outcomes and the magnitudes of their changes.

Furthermore, the literature has primarily favored simplistic single-component methodologies. While some studies have delved into the economic interdependence by

examining the cost-effectiveness of concurrent maintenance across multiple components, there remains a notable gap in addressing both stochastic dependence, wherein the failure of one component impacts the performance of other system elements, and structural dependence, pertaining to maintenance interconnections among various parts. Future research should prioritize the development of comprehensive modeling frameworks that encompass economic, stochastic, and structural dependencies. Embracing such a holistic perspective holds the promise of unlocking deeper insights into multicomponent railway systems. This, in turn, would also enrich the value of complex fault diagnosis tasks by enhancing capabilities in isolating specific fault types and pinpointing interdependent occurrences.

To bypass the challenges associated with uncertainty, explainability, and multicomponent dependence, researchers have resorted to using highly specific labeled data to train supervised models capable of recognizing fault patterns, their locations, and impacts. However, while this approach offers a simplified workaround, labeling data with such specificity can indeed be a complex and resource-intensive endeavor, particularly in real-world railway systems where faults occur infrequently and under varying conditions. This practical challenge is further compounded by the presence of large volumes of data with imbalanced distributions. Therefore, future research should prioritize strategies aimed at overcoming the limited availability of labeled inputs post-deployment and automating data labeling methods. Four lines of research have been overlooked in this regard and warrant further exploration: multimodal data fusion, transfer learning, semi-supervised learning, and active learning.

For instance, while this survey primarily focuses on time series data, no specific exclusion criteria were set regarding the inclusion of other data types. However, it is notable that there is a surprising scarcity of works integrating contextual data from diverse sources such as failure records, equipment logs, maintenance history, and traffic images. This omission highlights the potential of advanced techniques in multimodal data fusion. By harnessing contextual information for automatic labeling, such efforts could

effectively reduce manual efforts and enhance efficiency in model training, particularly when combined with sensor data.

Besides leveraging knowledge and labels from related modalities, another research direction lies in learning from related domains. This is particularly advantageous in the wheel-track interface, composed of a wide array of essential components such as wheels, axle bearings, tracks, joints, turnouts, rolling bearings, ballasts, and fasteners. This interdependent structural composition suggests that certain fault patterns and characteristics may be consistent across various railway systems or similar monitoring objects within the same system. Transfer learning techniques can effectively harness this shared structure and underlying patterns by transferring knowledge from pre-trained models or datasets in related domains to bootstrap model training. Nonetheless, further research is necessary to explore the effective transfer of knowledge within this diverse composition, while ensuring model generalization across different railway conditions and environments, encompassing variations in wagon types, track geometries, wheel geometries, and other relevant factors.

In addition to integrating new sources of information to alleviate limited labels, a promising but overlooked alternative is to address the labeling bottleneck itself. This can be achieved by selectively querying the labels of unlabeled instances to an oracle, prioritizing instances that are most uncertain, such as those near the discriminative hyperplane, and whose labels would provide the greatest value to the learning process. Future research should focus on developing active learning strategies to minimize the cost of collecting labeled data.

Lastly, instead of worrying about limited data, one can focus on exploiting the information contained in unlabeled data points. In this regard, semi-supervised techniques are extremely useful in addressing issues related to model updating and retraining in limited labeled data environments, despite not addressing the primary question of active learning, which is when to query. Future research efforts should focus on semi-supervised learning strategies, such as self-training and co-training, to capitalize on the potential of unlabeled data and further improve model performance.

Concerning the labeling process itself, the literature has been following the annotation scheme in Fig. 4. In RUL estimation, by counting backwards from a known moment of failure, stopping when reaching the previous one. In failure prediction, by considering a fixed window of instances before the failure as positive examples. In fault diagnosis, by considering solely the moment of failure as a positive example.

However, considering most of the mechanical components in the wheel-track interface present a slow gradual

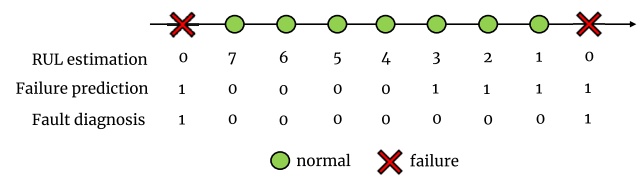


Fig. 4 Traditional annotation scheme, where green circles represent normal behavior and red crosses represent failures

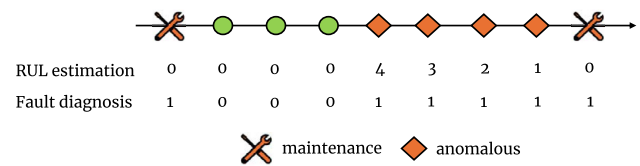


Fig. 5 Proposed annotation scheme, where orange squares represent anomalous behavior and green tools represent maintenance interventions

degradation process, failures might be months apart. With tens of thousands of observations in between, it is neither intuitive nor useful to create the labels at such a magnitude. Moreover, the moments of failure are not always known. Indeed, a maintenance inspection happens before a failure occurs, which shadows the real failure moment. To alleviate these issues, future research could lie in merging both fault diagnosis and prognosis techniques. The RUL backward count could be limited upon reaching the first anomalous observation diagnosed since the previous failure, as illustrated in Fig. 5. The first anomalous state would be automatically determined by the system's diagnosed degradation exceeding and maintaining above a tolerance threshold.

Indeed, the assumption of unlimited labeled data represents a fundamental challenge that undermines the robustness of model performance. Moreover, while much of the surveyed literature relies heavily on hand-engineered features tailored specifically for time series analysis, the lack of robustness in these features further compounds the issue. Recent efforts exploring novel deep learning architectures have underscored the shortcomings of traditional hand-crafted features, revealing the potential of feature representation learning to significantly enhance model performance across various tasks. The next natural step lies in meta-learning, especially within neural networks. By integrating joint feature, model, and algorithm learning, future research should target replacing prior hand-designed learners by learned learning algorithms. In fact, meta-learning would not only serve as a development tool but also provide the ability to update and maintain the model. By building a database of historical experiments and their outcomes, meta-learning algorithms can quickly identify

promising algorithms and hyperparameters, accelerating the optimization process.

In addition to meta-learning, various automated machine learning (AutoML) techniques present promising avenues for breaking the strong barrier between the training and testing phase, such as metaheuristics and Bayesian optimization. While data preprocessing and feature engineering are acknowledged as crucial for effective time series data mining in the surveyed literature, automating these processes remains challenging due to their inherent lack of robustness and the need of domain-specific factors. Therefore, priority should be given to the development of combined algorithm selection and hyperparameter tuning (CASH) techniques tailored explicitly to the unique demands of railway maintenance applications.

While these methodologies show promise in railway systems that accumulate knowledge over time, retaining all historical data remains highly impractical. Conversely, updating parameters solely based on recent data introduces a bias toward that data. Indeed, the existing literature predominantly favors traditional approaches, heavily reliant on the assumption of stationarity, a notion frequently invalidated by real-world environmental and operational variations. Despite some proposed solutions, there exists a pressing demand for memory-constrained algorithms capable of inherently adapting to current concepts while retaining past knowledge. This highlights the necessity for incremental algorithms incorporating mechanisms for both retention and forgetting. For instance, retention could be achieved through deep generative replay, employing generative models to encapsulate summaries of previous concepts, while forgetting could be realized through adaptive windowing techniques dynamically adjusting window lengths based on real-time monitoring of environmental cues, performance indicators, or distributional changes.

Despite these incremental strategies offering promise in managing the vast data influx within railway environments, they may still fall short. The aggregation of extensive time series data from sensors spread across extensive rail networks demands a nuanced equilibrium between memory constraints and predictive performance. This trade-off underscores the need for scalable algorithms as a focal point of research, by distributing training data or model components across multiple processing units, while synchronizing parameter updates.

Finally, to ensure the successful translation of research findings into practical applications, it's essential to conduct thorough benchmarking while considering the various factors discussed. Future research should include evaluation of performance metrics, while addressing cost considerations, interpretability, and computational expenses. Moreover, active engagement of key stakeholders, including maintenance staff and management, is crucial at every

developmental stage to ensure widespread acceptance of these systems.

8 Conclusions

In light of four essential requirements for creating an effective maintenance plan, this survey has examined the significant progress in low-cost and easily maintainable condition monitoring systems on railways. Regarding the suitability of employing a condition-based maintenance approach, our findings affirm that this strategy proves most effective in addressing the gradual deterioration processes observed at the wheel-track interface, such as wear and rolling contact fatigue. This interface stands out as the primary focus in railway monitoring. With respect to data availability, this survey has prioritized time series data due to its ubiquity in sensor-generated information and the inherent complexity associated with its temporal structure. Within this context, the following suggestions for future research directions (*RD*) are presented:

- (*RD1*) Formulating the problem in detail to properly select effective preprocessing techniques for dealing with the temporal dimension, as well as determining the modeling strategy.
- (*RD2*) Quantifying model uncertainty as pointwise evaluation measures and explanations of the model's predictions, leveraging the surveyed techniques on probabilistic modeling and Bayesian methods.
- (*RD3*) Computing multiple model explanations over time, with attribute-value contributions for prediction outcomes and the magnitudes of their changes, through interpretable models or post hoc explanation techniques.
- (*RD4*) Adopting more comprehensive multicomponent modeling approaches, that simultaneously encompass economic, stochastic, and structural dependencies.
- (*RD5*) Automating data labeling methods for large volumes of sensor data while addressing the limited availability of labeled inputs after model deployment.
- (*RD6*) Integrating contextual data on failures, ballast history, traffic images, inspections, and interventions data, with multimodal data fusion techniques.
- (*RD7*) Exploring effective transfer learning strategies in shared fault patterns and structural characteristics to bootstrap model training, while ensuring generalization across different railway conditions and environments.
- (*RD8*) Developing active learning strategies, by selectively querying the labels of unlabeled instances to an oracle, prioritizing instances that are most uncertain.

- (RD9) Exploiting the information contained in unlabeled data points, with semi-supervised learning strategies, such as self-training and co-training.
- (RD10) Adopting new labeling processes that consider both the slow gradual degradation process and unknown moments of failure, by complementing fault diagnosis and prognosis techniques.
- (RD11) Integrating joint feature, model, and algorithm learning, especially in neural networks, with an additional meta-learning layer.
- (RD12) Automating algorithm selection and hyperparameter tuning with AutoML techniques, such as meta-learning, metaheuristics, and Bayesian optimization.
- (RD13) Developing incremental memory-constrained algorithms with both remembering and forgetting mechanisms to inherently adapt to current concepts while preserving past knowledge.
- (RD14) Developing scalable algorithms, that accommodate increased data volume, by distributing training data or model components across multiple processing units and synchronizing parameter updates.
- (RD15) Including evaluation of performance metrics, while addressing cost considerations, interpretability, and computational expenses. Moreover, active engagement of key stakeholders is key.

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Data availability The authors declare that the data supporting the findings of this study are available within the paper.

Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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